

Q1

$$f(x) = \frac{x}{x+1}, \quad x \in \mathbb{R} - \{-1\}$$

(a) domain, range of f

$$\text{Domain} : \mathbb{R} - \{-1\}$$

$$\text{Range} : \mathbb{R} - \{-1\}$$

(b) One to one

$$f(x_1) = f(x_2)$$

$$\frac{x_1}{x_1+1} = \frac{x_2}{x_2+1}$$

$$x_1 = x_2$$

(c) onto

$$\text{let } y = \frac{x}{x+1}$$

$$x = \frac{y}{1-y}, \quad y \neq 1 \rightarrow \text{onto } \mathbb{R} - \{-1\}$$

(d) inverse

$$y = \frac{x}{x+1}$$

$$x = \frac{y}{1-y}$$

$$f^{-1}(x) = \frac{x}{1-x}, \quad x \neq 1$$

Q2

$$F(x) = x^3 + 1$$

(a) domain, range $= \mathbb{R}$

(b) Bijective
strictly increasing
onto
one to one

(c) even, odd or neither.

$$F(-x) = (-x)^3 + 1$$

$$= -x^3 + 1$$

$$\neq F(x)$$

$$\neq -F(x)$$

$\therefore$ neither.

Q3

$$F(x) = |x|$$

(a) domain, range.

$\downarrow$
 $\mathbb{R}$

$\downarrow$
 $[0, \infty)$

(b) even, odd, neither.

$$F(-x) = |-x| = |x| = F(x)$$

$\therefore$ even.

(c) Injective $\rightarrow \times$

$$F(2) = F(-2) = 2$$

but $2 \neq -2$

Q4

$$f(x) = \sin x$$

(a) domain, range.

 $\downarrow$
 $\mathbb{R}$ $\downarrow$
 $[-1, 1]$

(b) periodic, even, odd, neither

$$f(-x) = -\sin(x) = -f(x) \rightarrow \text{odd}$$

periodic $\checkmark \rightarrow 2\pi$ $\checkmark$ (c) many to one $\checkmark$

$$\sin\left(\frac{\pi}{6}\right) = \sin\left(\frac{5\pi}{6}\right)$$

Q5

(i) $f(x) = x^2$

many to one $x^2 = (-x)^2$ not onto (negatives missing)
even.

(ii) $\log_2 x$

one to one

not onto ($y \in \mathbb{R}$)

neither even nor odd

(iii) $2x - 3$

one to one

onto

odd $\rightarrow f(-x) = -f(x)$

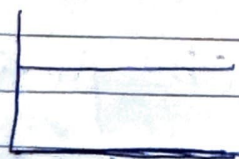
Q9 $f(-x) = f(x)$, even, $f(x) = x^2$, $f(x) = \cos(x)$
 Q10 $f(-x) = -f(x)$, odd, $f(x) = x^3$, $f(x) = \sin(x)$

(iv) $|x|$

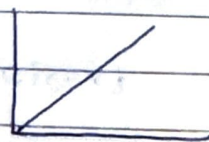
many to one
 not onto
 even.

cl6

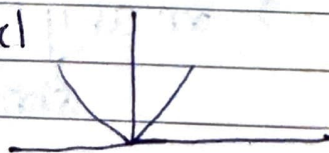
Constant $f(x) = 5$



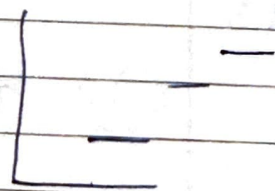
Identity $f(x) = x$



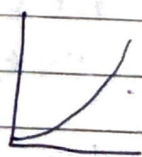
Modulus $f(x) = |x|$



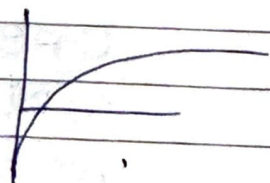
greatest int. $f(x) = [x]$



expo. $f(x) = e^x$



Log. $f(x) = \log(x)$



Q7 $F(x, y) = x^2 + y^2$

(a) Domain $\mathbb{R}^2$

Range $[0, \infty)$

(b) $x^2 + y^2 = C$
 circles.

(c) Surface: paraboloid $z = x^2 + y^2$

cl8

$F(x, y) = \sqrt{y - x^2}$ Domain $(x, y) \in \mathbb{R}^2$
 $y \geq x^2$

Not defined for all $\mathbb{R}^2$ ($y < x^2$) Range $[0, \infty)$